- If a magnet is broken into number of pieces, each piece becomes an independent magnet which has North and South.
- A magnet can impart its properties by induction to any magnetic material.

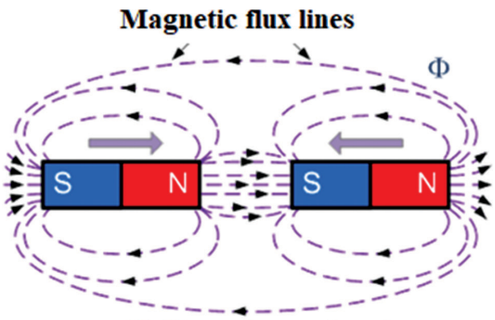


Fig 3.2 *UnLike Poles*

- A magnet loses its properties when it is heated, hammered or dropped from height.

3.2.1 Classification of Magnets

1. Natural Magnet

The magnet found in nature is known as lodestone. The natural magnet is one of the iron ore magnetite with chemical composition Fe_3O_4

2. Artificial Magnet

The magnets prepared by artificial method are called artificial magnets. It can be made in different shape, size and strength only in certain metals. There are two types of artificial magnet.

a) Permanent Magnet

In a permanent magnet, the magnetic materials can retain magnetic property permanently for a long time. Bar magnet, Horse Shoe magnet, Ring Magnet, Cylindrical magnet are some types based on shapes.



Fig 3.3 *Permanent Magnets*

ALNICO (Aluminium-Nickel-Cobalt) is an alloy metal specifically used as permanent magnet because it can be lifted up to 50 times weight load compared to its own weight. Permanent magnets can be formed by touch method, electric current method and induction method.

b) Temporary Magnet (or) Electro Magnet

When an electric current is passed through a coil of wire wrapped around a soft iron core, a very strong magnetic field is produced.

This is called as electro magnet. If the current is cut off, the core will be demagnetized, and hence known as temporary magnet.

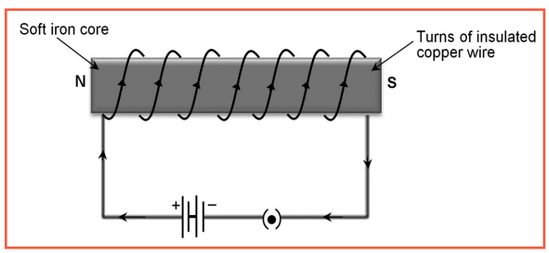


Fig 3.4 *Electro magnet*

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